

Lab companion — Federated Query with BigQuery and AlloyDB

Day 2 (Tue Oct 6), 10:45–11:15. The lab's own step-by-step lives in your Google Skills classroom seat (listed there as "Federated Query with BigQuery"); this companion is the orientation around it — what you're building, why, and what tends to go wrong.

The one thing to remember

A federated query is BigQuery's version of a linked server with OPENQUERY.

- The **connection resource** = the **linked server definition**: server, credentials, one-time setup.
- **EXTERNAL_QUERY(connection_id, '...')** = **OPENQUERY**: the string inside the quotes is shipped to the remote system and executed *there*, in *its* SQL dialect.
- And just like OPENQUERY: there is no browsing the remote tables — you query blind.

What you'll do

- Connect BigQuery to an AlloyDB for PostgreSQL database (AlloyDB plays the role of a live operational store — it is not your migration target).
- Run your first EXTERNAL_QUERY against it.
- Join historical analytical data (in BigQuery) with live transactional rows (in AlloyDB) in a single SQL statement.

Why it matters for your migration

During your migration, not everything moves at once. Federation means the SQL Server systems that are still operational stay reachable from BigQuery **without copying them first** — no extract job built just to join operational tables against warehouse data. This is the pattern that lets you migrate in phases instead of betting on a big-bang cutover.

The honest trade-offs, so you apply it deliberately: the source system takes the query load, results don't get BigQuery storage optimizations, federated queries are read-only, and on-demand billing charges for the bytes returned from the external side. Federation is a bridge, not a destination — Silver/Gold still belong in managed tables.

Before you start

- Launch the lab from your seat in the [Skills classroom](#) and sign in to the console with the **lab-provided credentials**, not your own account.
- Check the **project picker** at the top of the console: the lab spins up its own temporary GCP project, and everything you build must go there.
- Note the region the lab instructions give you, and keep every resource you create in that region.

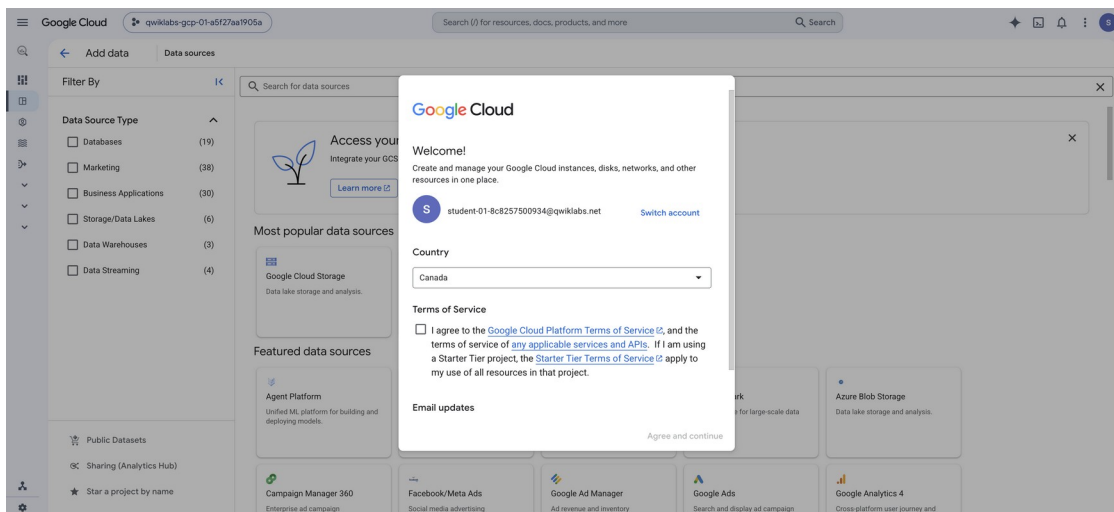
While you work — pointers, not steps

- **The connection setup is the 'linked server definition' moment.** New concept, one-time work: region, connection ID, credentials. Go slowly here; once it exists, everything else is just SQL.
- **API-enable and consent prompts are expected.** If the console asks you to enable an API (for example the BigQuery Connection API) or confirm a consent screen, click through — that's normal in a fresh lab project.
- **Your first EXTERNAL_QUERY:** the inner statement is **PostgreSQL**, executed on the AlloyDB side. BigQuery syntax inside the quotes fails — this is the single most common trip point. Watch your quoting too: single quotes inside the inner query must be escaped (double them, or wrap the whole inner query in triple single quotes `'''...'''` the way Google's own examples do).
- **The payoff is the join:** historical/analytical data on the BigQuery side live transactional rows on the AlloyDB side, one statement. If you get nothing else working, get this — a working connection you can re-create later is the real takeaway.

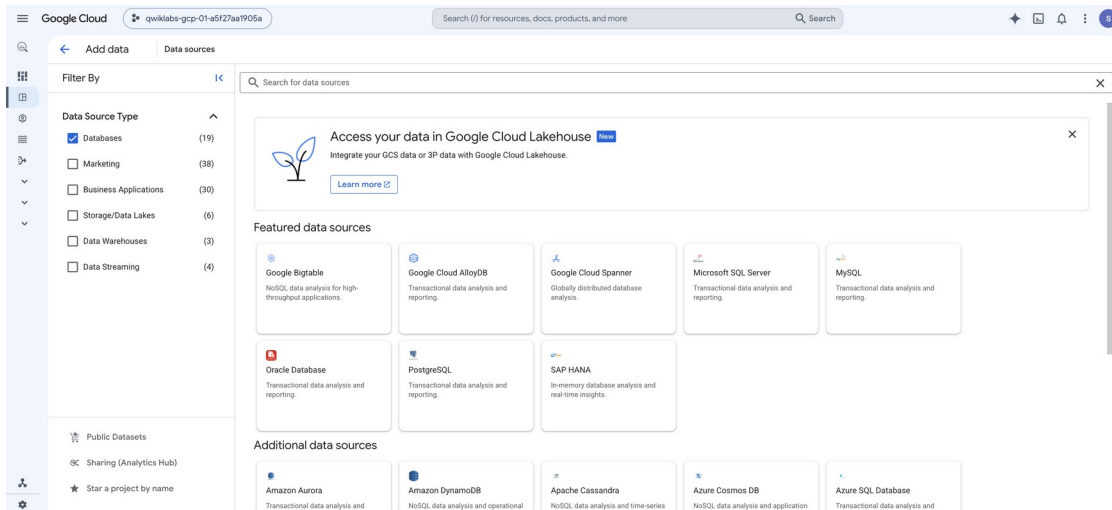
The critical steps, in pictures (recorded from a real run)

These frames are from an instructor dry-run of this exact lab. Project IDs, student logins and passwords in them are per-lab — yours will differ; the steps are identical.

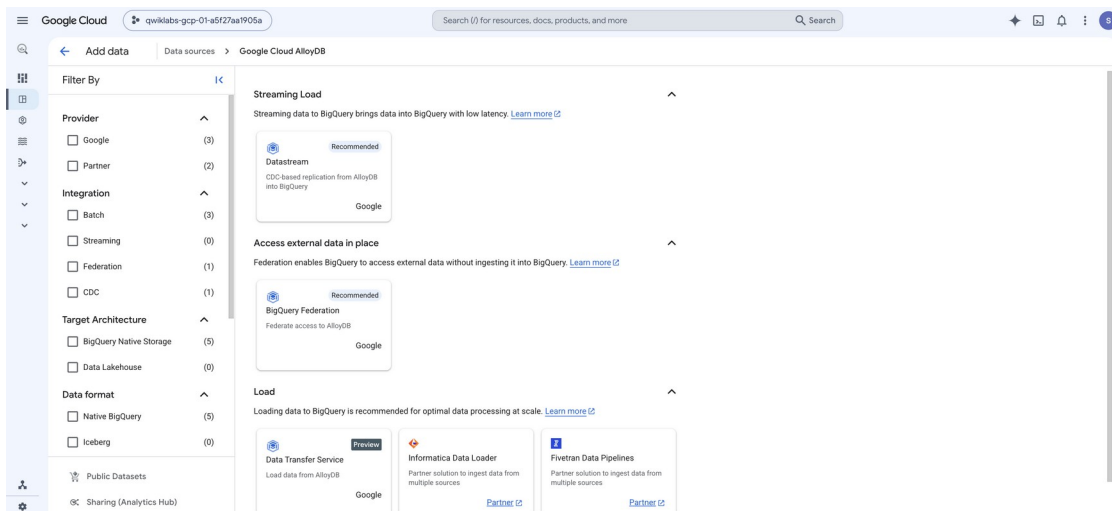
Consent and API prompts are part of the ritual in a fresh lab project — click through them.



BigQuery ► Studio ► ♦ Add data ► Data Source Type = **Databases** ► the **Google Cloud AlloyDB** card.



Pick **BigQuery Federation** under "Access external data in place" — federation, not a load or a Datastream replication.



The "linked server definition" moment: Connection type **AlloyDB**, Connection ID **AlloyDB-weblog**, Location type **Region**, region **us-central1**. Go slowly here — everything after this is just SQL.

Google Cloud | qwiklabs-gcp-01-a5f27aa1905a | Search (G) for resources, docs, products, and more

← Add data | Data sources | Google Cloud AlloyDB

Filter By

- Provider
 - Google (3)
 - Partner (2)
- Integration
 - Batch (3)
 - Streaming (0)
 - Federation (1)
 - CDC (1)
- Target Architecture
 - BigQuery Native Storage (5)
 - Data Lakehouse (0)
- Data format
 - Native BigQuery (5)
 - Iceberg (0)

Public Datasets | SC Sharing (Analytics Hub)

Streaming Load
Streaming data to BigQuery brings data into BigQuery with low latency. [Learn more](#)

Access external data in place
Federation enables BigQuery to access external data without ingesting it into BigQuery. [Learn more](#)

Load
Loading data to BigQuery is recommended for optimal data processing at scale. [Learn more](#)

External data source

Connection type: AlloyDB

Connection ID: AlloyDB-weblog

Location type: Region
Specify a region to colocate your datasets with other Google Cloud services.

Multi-region: Allow BigQuery to select a region within a group to achieve higher quota limits.

Some locations have been restricted due to a policy set by your organization. [Learn more about restricting locations.](#)

Region: us-central1 (Iowa)

Friendly name

Description

Encryption: Google-managed encryption key
Keys owned by Google

Cloud KMS key: Keys owned by customers

Username: postgres

Password: [masked]

Database: postgres

AlloyDB instance: //alloydb.googleapis.com/projects/qwiklabs-gcp-01-a5f27aa1905a/locations

Create connection | Cancel

Bottom half of the same form: username postgres, the password from **your** lab's instructions panel, database postgres, and the full AlloyDB instance path — then **Create connection**.

Google Cloud | qwiklabs-gcp-01-a5f27aa1905a | Search (G) for resources, docs, products, and more

← Add data | Data sources | Google Cloud AlloyDB

Filter By

- Target Architecture
 - BigQuery Native Storage (5)
 - Data Lakehouse (0)
- Data format
 - Native BigQuery (5)
 - Iceberg (0)
 - Delta (0)
 - Hive (0)
- Engine
 - Data Transfer Service (1)
 - Dataflow (0)
 - Datastream (1)
 - BigQuery Federation (1)
 - Pub/Sub (0)

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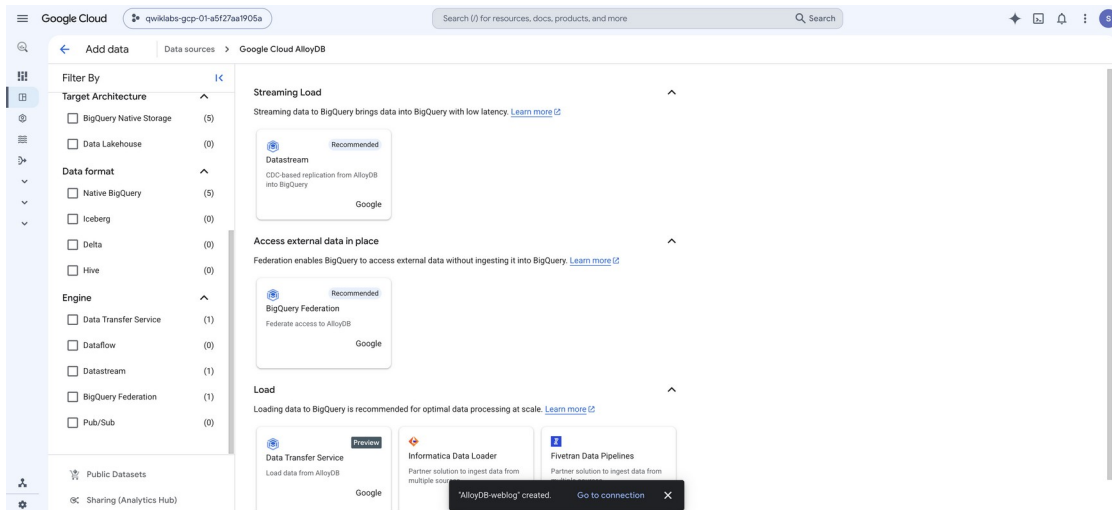
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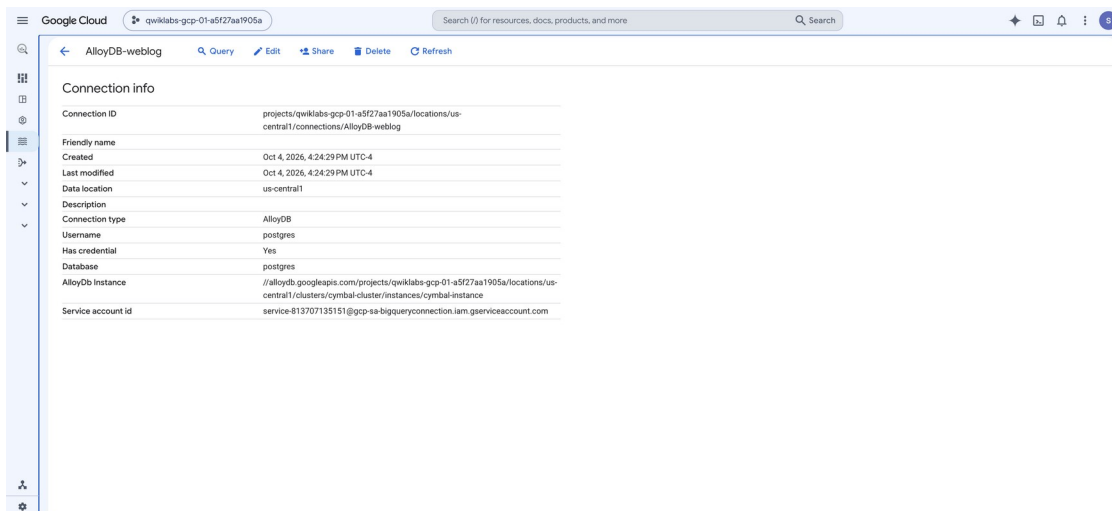
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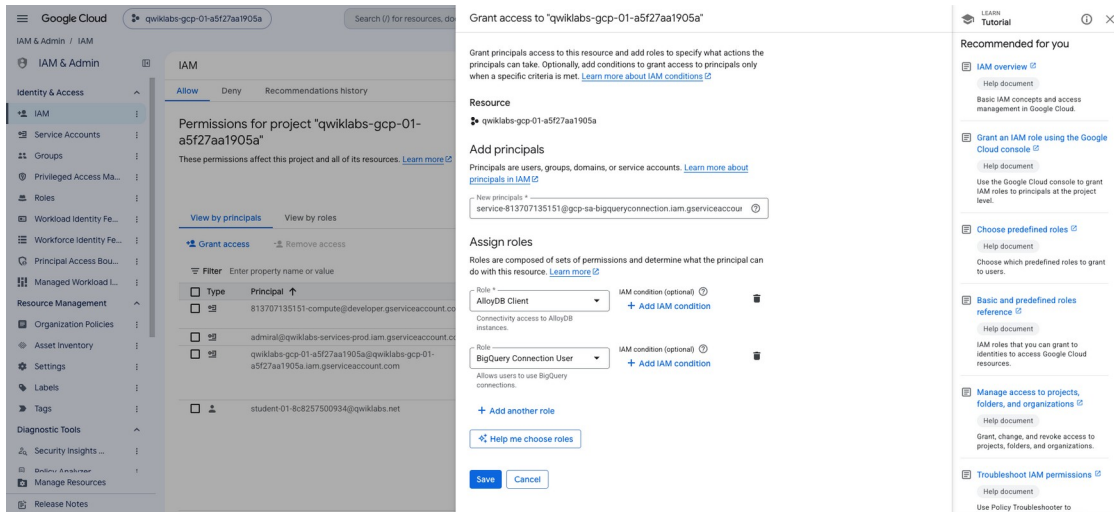
"AlloyDB-weblog" created. The linked server exists; nothing in it can be queried yet — its service account needs IAM first.



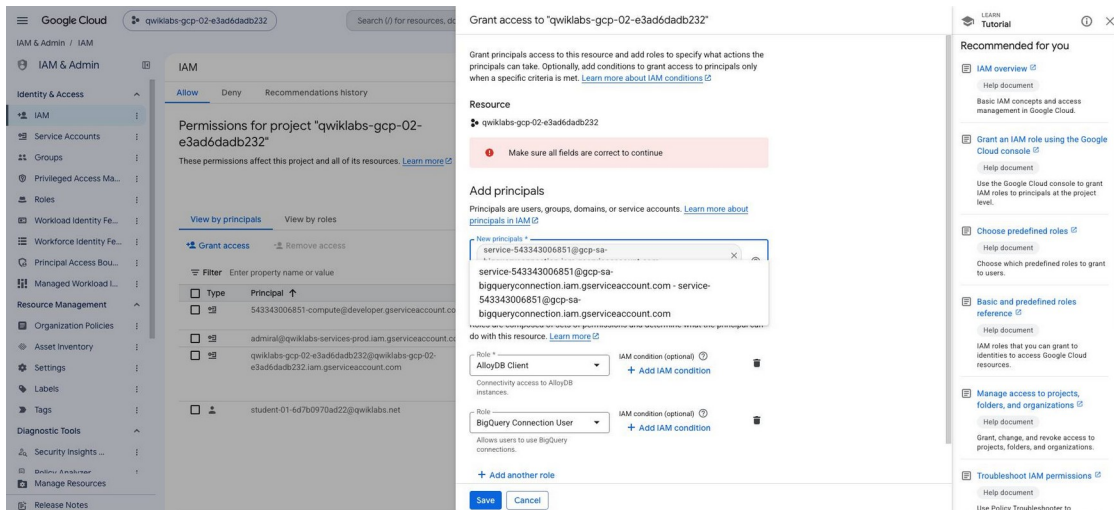
Open the connection (Pipelines & Connections ► Connections ► AlloyDB-weblog) and copy the **Service account id** — this is the principal the IAM grant goes to.



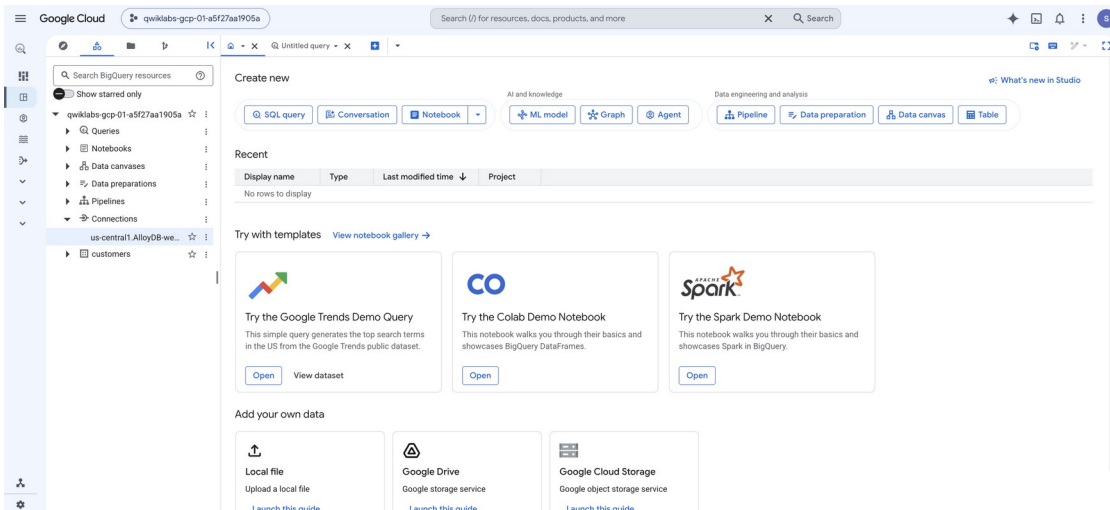
IAM ► **Grant access**: paste the service account, **press ENTER to make it a chip**, then add both roles — **AlloyDB Client** and **BigQuery Connection User** — and Save. Wait for the "Policy updated" toast.



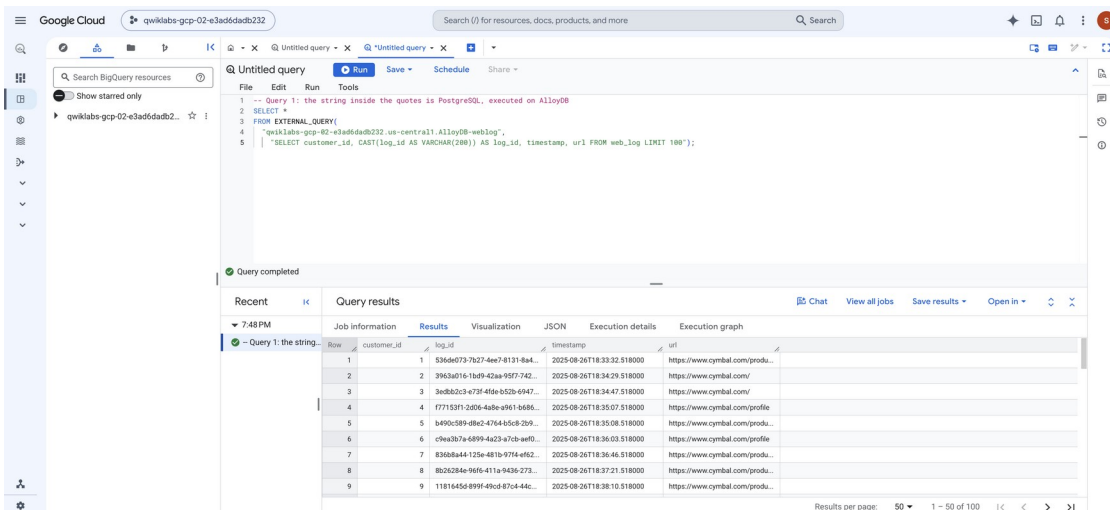
What the chip trap looks like: the email is in the box, yet Save says "Enter at least one principal" — the text was never committed as a chip.



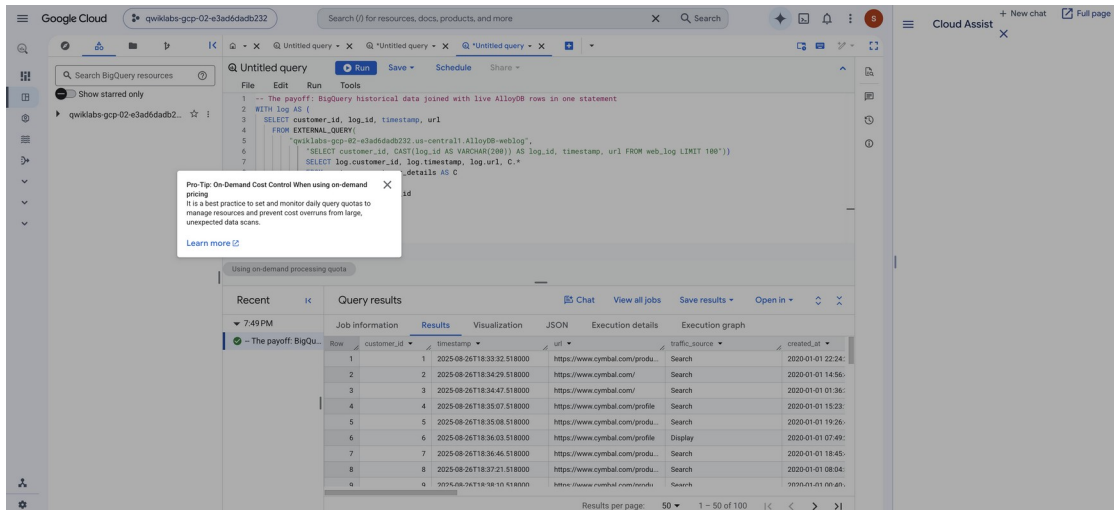
Self-check landmark: the connection shows under **Connections** in the Explorer tree, next to the customers dataset — remember, the AlloyDB tables will NOT appear here; federation is query-only.



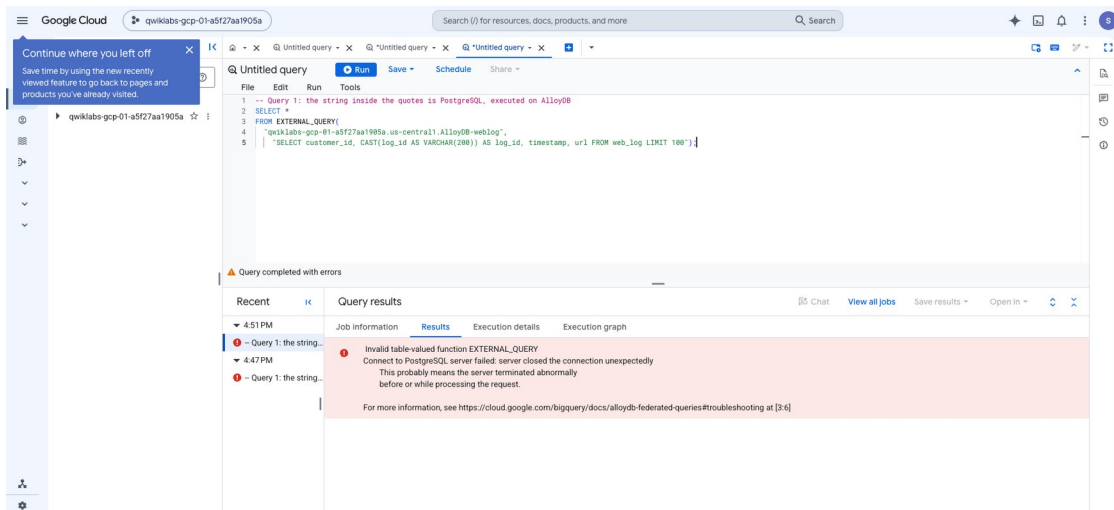
First `EXTERNAL_QUERY`, green: rows coming back from AlloyDB's `web_log` while you sit in BigQuery. The inner string ran as PostgreSQL on the AlloyDB side.



The payoff — one statement, two homes: `url/timestamp` from live AlloyDB rows joined with `traffic_source`, `zip` and `state` from the BigQuery `customers` table. If you get nothing else working in the lab, get this.



And when it goes wrong, this is the face of it — the PostgreSQL connection error with Google's own troubleshooting link. Nine times out of ten it is the IAM chip trap or the per-lab password.



Common gotchas

- **'Connect to PostgreSQL server failed: server closed the connection unexpectedly'** — this one error has two sneaky causes, both at setup time. First, credentials: the connection form values are per-lab — copy the password from YOUR running lab's instructions panel (it matches your console password), not from a stale manual page or a previous run. Second — and this one bites everyone — the IAM grant silently never happened: see the chip trap next.
- **The principal chip trap (the #1 silent failure):** when you paste the connection service account into "New principals" on the IAM "Grant access" panel, you MUST press Enter afterwards so the address turns into a chip — otherwise Save is rejected with "Enter at least one principal" (easy to miss: the email is sitting right there in the box). If you click a suggestion as well, you can end up with a duplicate chip — delete one with its ×. The grant only counts when you see the "Policy updated" toast; check the

principal actually appears in the IAM list. Without this grant, every `EXTERNAL_QUERY` fails with the PostgreSQL connection error above.

- **Wrong project in the picker** — you created the connection in the wrong project and now nothing in the lab matches. Check the picker first whenever something "isn't there."
- **Connection ID typos in `EXTERNAL_QUERY`** — it must match exactly, in the form `region.connection-id` (for example `'us.my-connection'`).
- **Region alignment** — the connection resource's location must match where the query runs, and a single-region BigQuery setup can only federate to sources in the same region. If the lab says one region, use it everywhere.
- **BigQuery SQL inside the quotes** — the inner string is PostgreSQL. Backticks, GoogleSQL functions, and T-SQL habits all fail in there.
- **'Why can't I see the AlloyDB tables in the Explorer pane?'** — because federation is query-only. There is no table browsing; you write the inner query blind, exactly like `OPENQUERY`.

If you finish early

- Run the inner PostgreSQL query on its own (just the string inside the quotes) to prove to yourself which side executes it.
- Add a filter inside the inner query — for example a `WHERE` or a `LIMIT` — and watch the remote side do the work before BigQuery sees a row.
- List the remote database's metadata through the same connection:

```
SELECT * FROM EXTERNAL_QUERY("region.connection-id",  
  "select * from information_schema.tables;");
```

(swap in your connection ID — this is the PostgreSQL `information_schema`, not BigQuery's).

- Open the job's **Execution details** after the join and find what BigQuery reports for the federated step.

Self-check

- ☐ I can see my connection resource under **External connections** in the BigQuery Explorer pane.
- ☐ My first `EXTERNAL_QUERY` returned rows from AlloyDB.
- ☐ I joined BigQuery historical data with live AlloyDB transactional rows in one query.
- ☐ I can explain which part of the query runs where: the inner string is PostgreSQL on AlloyDB; everything outside `EXTERNAL_QUERY` is GoogleSQL in BigQuery.
- ☐ I can name what this maps to in my current stack: a linked server definition plus `OPENQUERY`.

What this replaces in your current stack

- **Linked servers, four-part names, and OPENQUERY** — the connection resource plus EXTERNAL_QUERY covers the same ground.
- **The nightly 'extract the operational tables just so we can join them' jobs** — during the migration you query data where it lives and load it only when there's a reason to.

The debrief in one line: querying data where it lives — that is the lakehouse promise.

Your Bronze lands in GCS/BigQuery, but anything still operational on the SQL Server side stays reachable during the migration. You don't have to move it all first.

Dig deeper

- [Introduction to federated queries](#) — region rules, quotas, pricing, SQL pushdowns
 - [AlloyDB federated queries](#) — including the information_schema trick above
 - [AlloyDB overview](#) — what the lab's stand-in operational store actually is
 - Next: the same trick against files in Cloud Storage — [Exploring external tables](#)
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Part of the Iowa DOM DoIT Data Engineering student pack — see ../00_schedule.md.